# PHY2011-BIOPHYSICS LT3

# Module 1

- **Chemical binding and energetics:** Ionization Energy, Electron Affinity and
- Chemical Binding, Secondary Bonds, Interatomic Potentials for Strong and weak bonds, non-central forces,
- bond energies, spring constant
- **Energetics of Biological systems:** Internal energy,
- thermodynamics, entropy, equilibrium

# Health Informatics

- Health informatics is a multidisciplinary specialization dealing with the collection and management of health data and developing applications through computational data analysis to provide better healthcare strategies.
- The specialization bridges engineering and medicine.

Dr.M.Xavier Suresh

# What is Biophysics?

It is neither “physics for biologists”, nor “physical methods applied to biology”

It is a modern, interdisciplinary field of science leading to new approaches for our understanding of biological functions.

Mathematics +Physics +Biology + Chemistry

Paradigm: “Biological system is not simply the sum of its molecular components but is rather their functional integration” –example biological membrane.

Scale from organism to single molecule
Time scales from years to femtoseconds ($10^{-15}$), 1/1000 ps

# What Do Biophysicists Do?

Biophysicists work to develop methods to overcome disease, eradicate global hunger, produce renewable energy sources, design cutting-edge technologies, and solve countless scientific mysteries. In short, biophysicists are at the forefront of solving age-old human problems as well as problems of the future.

### DATA ANALYSES AND STRUCTURE

The structure of DNA was solved in 1953 using biophysics, and this discovery was critical to showing how DNA serves as a blueprint for life.

Now we can read the sequences of DNA from thousands of humans and all varieties of living organisms. Biophysical techniques are also essential to the analysis of these vast quantities of data.

### MOLECULES IN MOTION

Biophysicists study how hormones move around the cell, and how cells communicate with each other. Using fluorescent tags, biophysicists have been able to make cells glow like a firefly under a microscope and learn about the cell's sophisticated internal transit system.

### NEUROSCIENCE

Biophysicists are building computer models called neural networks to model how the brain and nervous system work, leading to new understandings of how visual and auditory information is processed.

### BIOENGINEERING, NANOTECHNOLOGIES, BIOMATERIALS

Biophysics has also been critical to understanding biomechanics and applying this information to the design of better prosthetic limbs, and better nanomaterials for drug delivery.

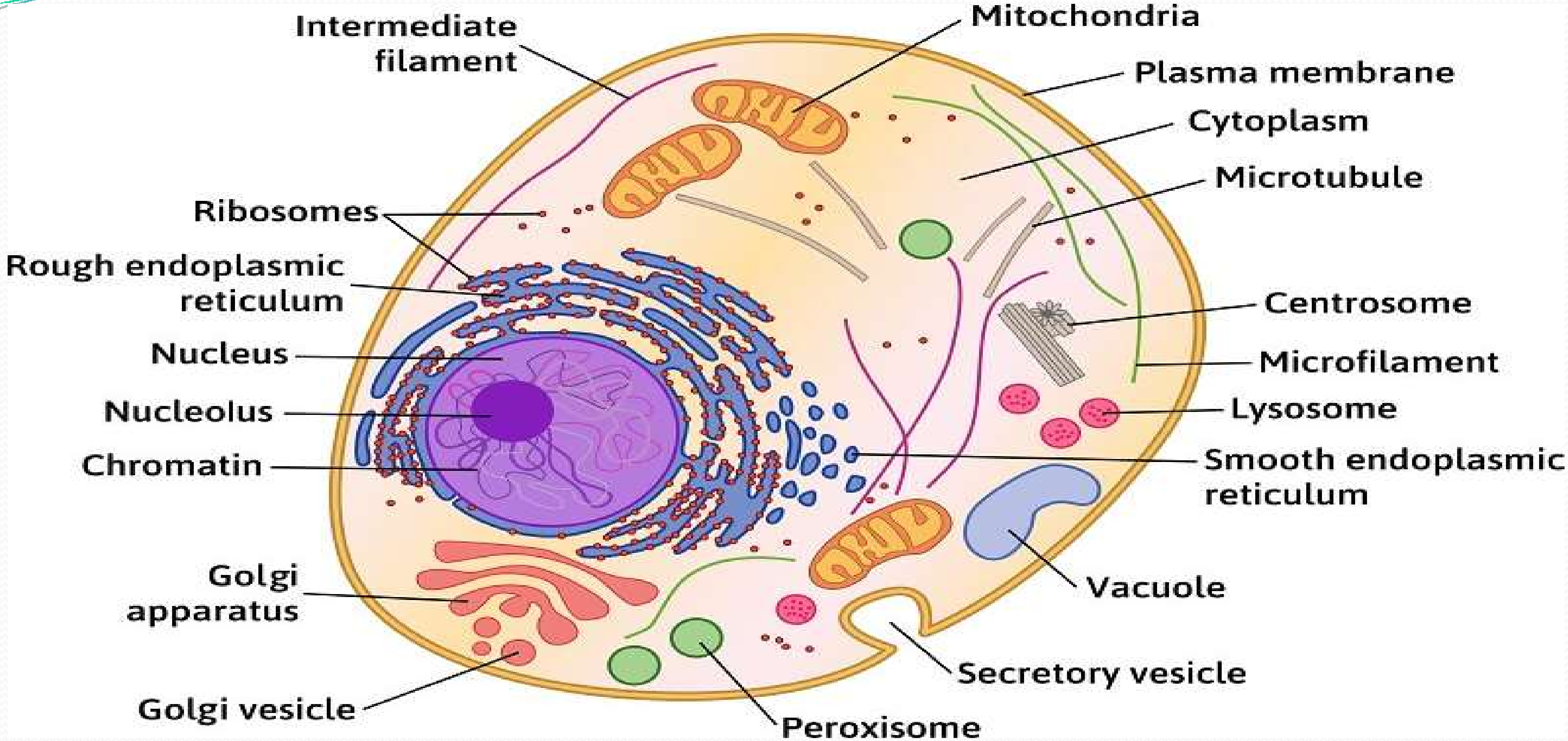
### COMPUTER MODELLING

Biophysicists develop and use computer modeling methods to see and manipulate the shapes and structures of proteins, viruses, and other complex molecules, providing crucial information needed to develop new drug targets, or understand how proteins mutate and cause tumors to grow.

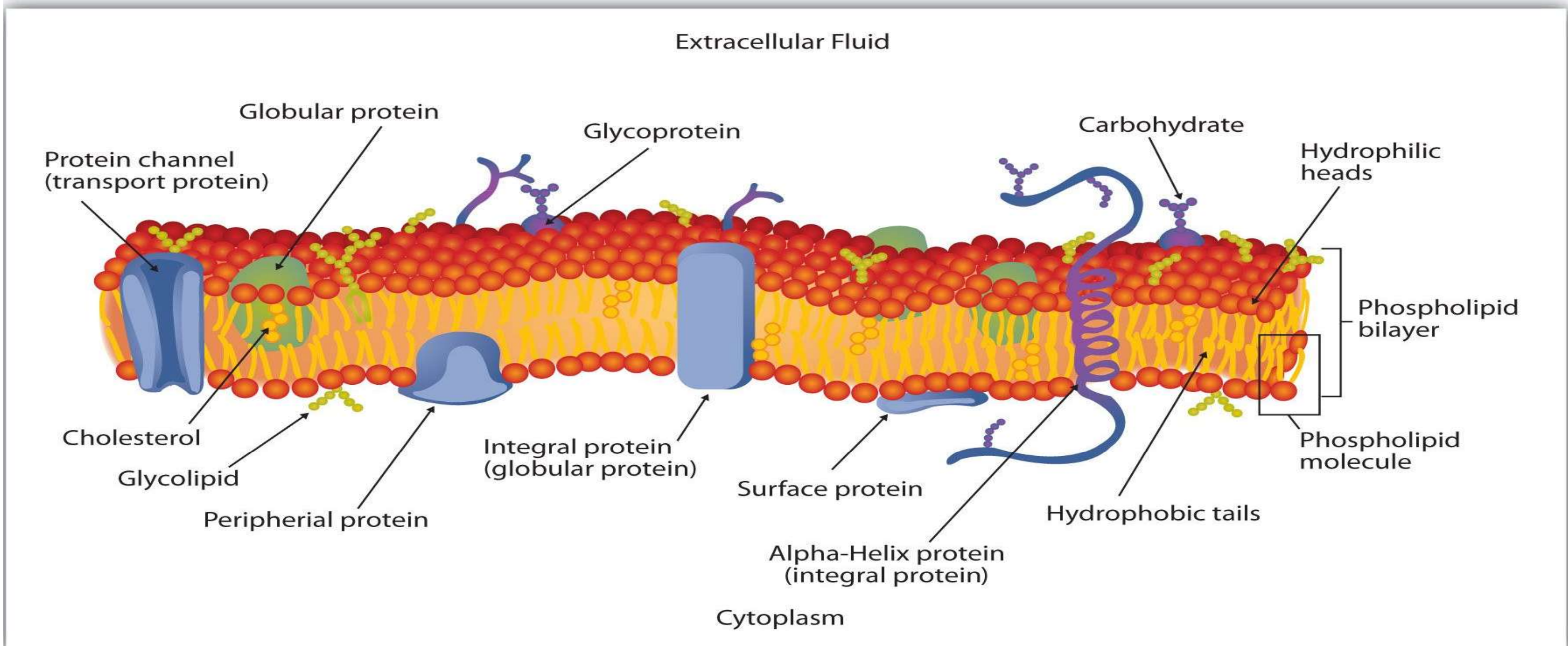
# What is Biophysics?

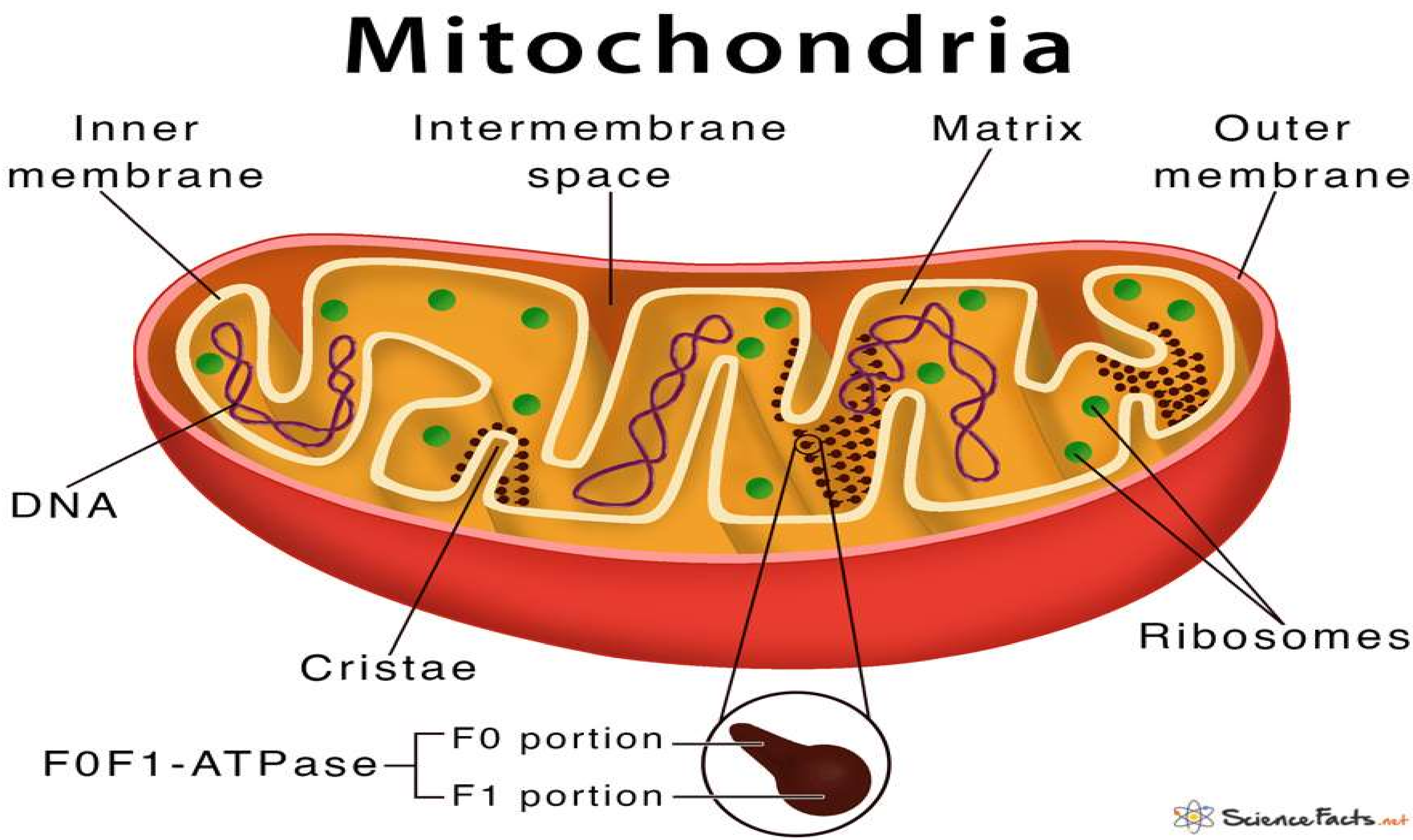
- Biophysics is that branch of knowledge that applies the principles of physics and chemistry and the methods of mathematical analysis and computer modeling to biological systems, with the ultimate goal of understanding at a fundamental level the structure, dynamics, interactions, and ultimately the function of biological systems.

# The Cell Structure

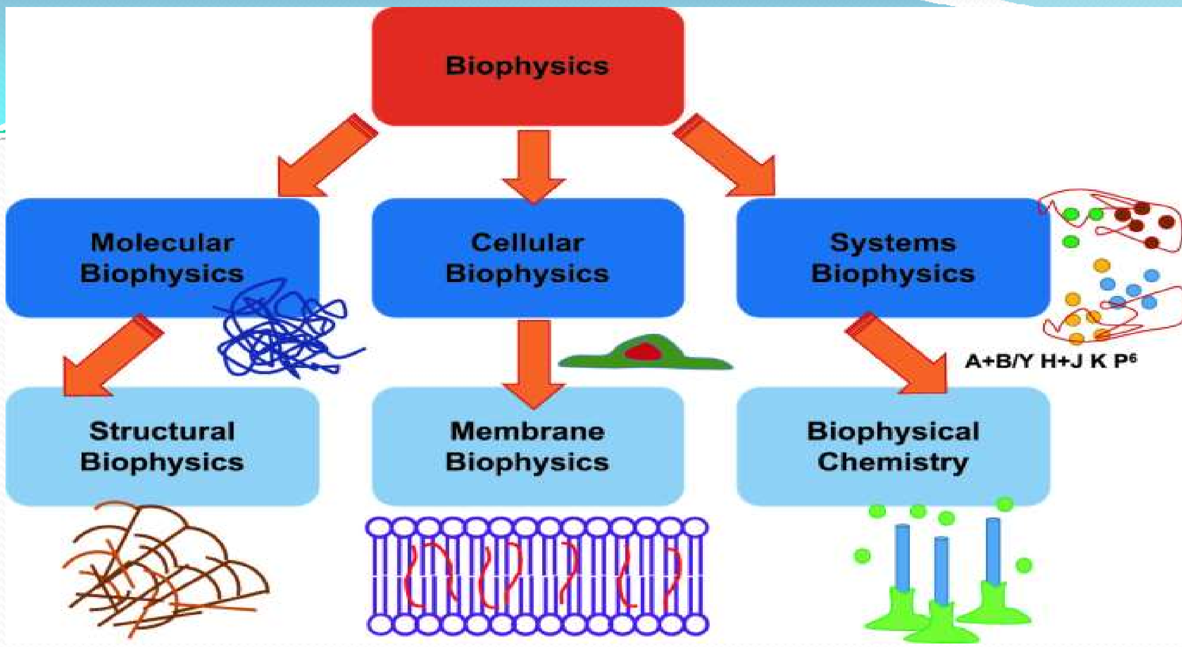


# Cell membrane




Mitochondria
Inner membrane
Intermembrane space
Matrix
Outer membrane
DNA
Cristae
Ribosomes
F0F1-ATPase
F0 portion
F1 portion
ScienceFacts.net

- The biological questions of interest to biophysics are as diverse as the organisms of biology:
- How do linear polymers of only 20 different amino acids fold into proteins with precise three-dimensional structures and specific biological functions?
- How does a single, enormously long DNA molecule untwist and exactly replicate itself during cell division?
- How does RNA fold into complex 3-D structures and carry out highly sophisticated transactions when it is composed of four chemically-similar nucleotides?
- How are sound waves, or photons, or odors, or flavors, or touches, detected by a sensory organ and converted into electrical impulses that provide the brain with information about the external world?
- How does a muscle cell convert the chemical energy of ATP hydrolysis into mechanical force and movement?
- How does the cell membrane, a lipid barrier impermeable to water-soluble molecules, selectively transport such molecules through its non-polar interior?
- Biophysics explains biological functions in terms of molecular mechanisms: precise physical descriptions of how individual molecules work together like tiny “nanomachines” to produce specific biological functions.




- Divisions of Biophysics:
- Molecular biophysics
- subcellular biophysics
- Environmental biophysics
- Imaging biophysics
- Medical biophysics
- Biomechanics
- Energy flow and bioenergetics
- Membrane Biophysics
- Physiological and Anatomical Biophysics
- Sensory Biophysics
- Theoretical Biophysics
- Computational Biophysics

Job Market:
Universities
Industry
Medical Centers
Research Institutes
Government

Impact on biotechnology and medicine

**History**

# *First Biophysicists*

Heraclitus 5th century B.C. – earliest mechanistic theories of life processes, insight into dynamic.

**"Change is central to Universe".**

**"Logos is the fundamental order of all "on Nature" changes of objects with the flow of time"**

**"You can not step twice into the same river"**

Epicurus 3rd century B.C. – atom. Living organisms follow the same laws as non-living objects.

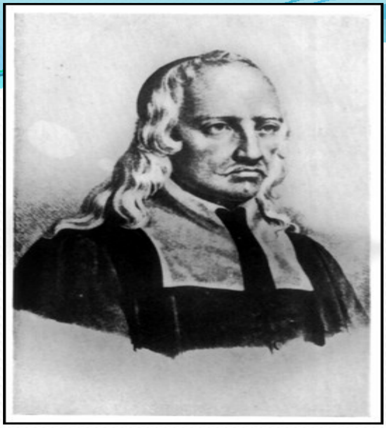
Galen 2th century AD – physician, most accomplished medical researcher of the Roman period. His theories dominated Western medicine for over millennium.

Better anatomy only by Vesalius in 1543
Better understanding of blood and heart in 1628

Leonardo da Vinci 16th century – mechanical principles of bird flight (to use for engineering design) – bionics

## History…

Borelli farther of biomechanics



Giovanni Alfonso Borelli 17th century- related animals to machines and utilized mathematics to prove his theories.

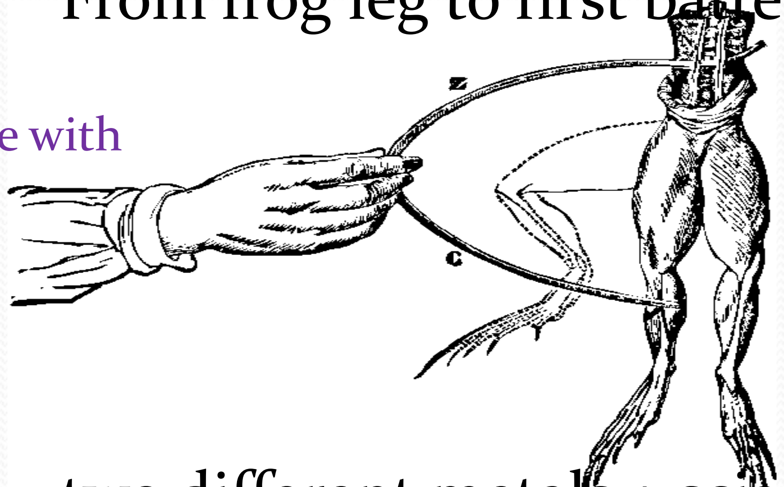
*De Motu Animalium* – comprehensive biomechanical description of limb's mobility, bird's flight, swimming movement, heart function.

History...

# Luigi Galvani / Alessandro Volta

From frog leg to first battery.

18th (1771) Galvani touched frog nerve with charged scalpel.




With two different metals effect is stronger.

Contact potential !!

Electric circuit = two different metals + sciatic nerve of the frog
Nerve of the frog's leg = electrolyte and sensor
Metals = electrodes
If close the circuit dead leg will twitch.

Volta created first battery by substituting frog leg with electrolyte.

- **Luigi Galvani**, an Italian physicist, discovered something he named, "animal electricity" when two different metals were connected in series with a frog's leg and to one another. Volta realised that the frog's leg served as both a conductor of electricity (what we would now call an electrolyte) and as a detector of electricity. He also understood that the frog's legs were irrelevant to the electric current, which was caused by the two differing metals. He replaced the frog's leg with brine-soaked paper, and detected the flow of electricity by other means familiar to him from his previous studies. In this way he discovered the electrochemical series, and the law that the electromotive force (emf) of **a galvanic cell, consisting of a pair of metal electrodes separated by electrolyte,** is the difference between their two electrode potentials (thus, two identical electrodes and a common electrolyte give zero net emf). This may be called Volta's Law of the electrochemical series.
- In 1800, as the result of a professional disagreement over the galvanic response advocated by Galvani, **Volta** invented the voltaic pile, an early electric battery, which produced a steady electric current. Volta had determined that the most effective pair of dissimilar metals to produce electricity was zinc and copper. Initially he experimented with individual cells in series, each cell being a wine goblet filled with brine into which the two dissimilar electrodes were dipped. The voltaic pile replaced the goblets with cardboard soaked in brine.

# History…

To get a better perspective, we consider a very brief history of physics.

**1700-1900 Classical era**

Mechanics (Newton)

Electromagnetism (Maxwell)

Thermodynamics/Statistical Mechanics

**1900-1950 Quantum era**

Relativity (Einstein)

Quantum Mechanics (Schroedinger, Heisenberg, Dirac)

Quantum Field Theory (Feynman, Schwinger, Tomonaga)

**1950-present** Modern era

Development of physics methods are complete.

Their applications to matter at all scales, however, are continuing and expanding.

## History…

**Paradigm shift in Molecular Biology**

A young science is always dominated by experimental work and is qualitative in its description rather than quantitative.

As it matures, theory/computations take a complementary role, which eventually results in the standard scientific approach:

**Exp. Data □ Model/theory □ Prediction □ Exp. test**

Thanks to increases in computing power, biomolecules can now be described near experimental accuracy.

Thus we are entering a quantitative phase in molecular biology where physical models will play crucial roles in further developments in biophysics.

# History

In 1892: Karl Pearson (missing link between biology and physics => name biophysics), The Grammar of Science

In 1943: Erwin Schrodinger (Nobel Prize, 1933), lecture series: What is Life

In 1946: Biophysics Research Unit, King's College, London, hire physicists to work on questions of biological significance; Maurice Wilkins, Rosalind Franklin: X-ray diffraction of DNA

1953: Francis Crick (particle physicist turned into biophysicist at Cambridge) and James Watson (biologist): double helix structure of DNA Watson and Crick – Double helix structure of DNA

- 1) DNA -> RNA -> protein,
- 2) Duplication problem

1957:The Biophysical Society founded

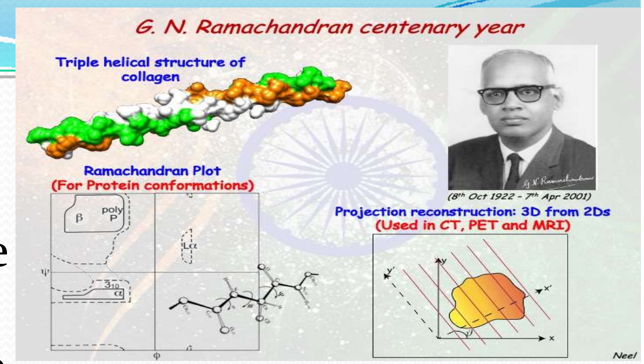
1959 Max Perutz and John Kendrew set about determining the structures of the oxygen transporting proteins myoglobin and its larger cousin haemoglobin. 1959

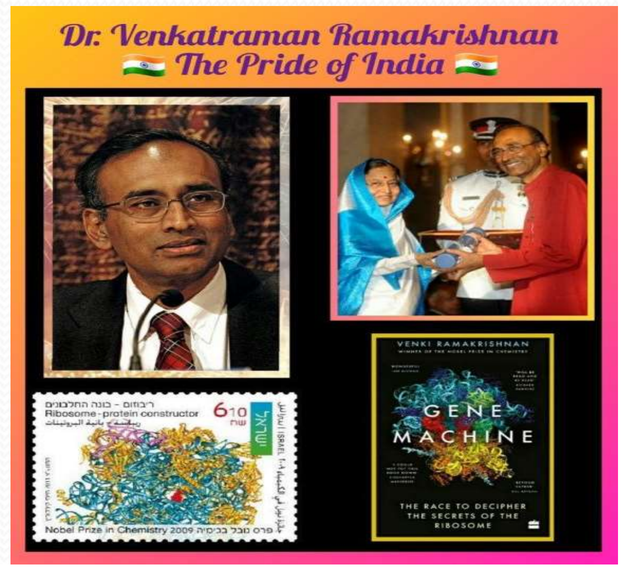
1960-present – Exponential growth in protein structure determination via X-ray and NMR methods (computers play an essential role)

# History…

- Structure determination by means of X-ray diffraction, this approach reached its zenith around the middle of the 20th century.
- 2001 Craig Venter – Complete reading of human genome
- Two major problems in molecular biology:
- • Protein folding (finding the tertiary structure given the primary seq.)
- • Molecular recognition and assembly .

- Within a decade, the secrets of these key structures had been exposed, important input having come from the knowledge acquired of the bonding between atoms, thanks to the efforts of physicists.
- GN Ramachandran 1954, triple helical structure of collagen, Ramachandran plot 1963
- Venkataraman Ramakrishnan 2009- structure and function of ribosome



- Schro¨dinger's time-independent equation

$$\mathcal{H} \cdot \Psi = \mathcal{E} \cdot \Psi$$

$$\mathcal{H} \cdot (C_1 \cdot \Psi_1 + C_2 \cdot \Psi_2) = \mathcal{E} \cdot (C_1 \cdot \Psi_1 + C_2 \cdot \Psi_2)$$

**•What happens when two atoms approach each other ?**

•The interatomic potential is repulsive at sufficiently short range and attractive at sufficiently long range, and the implication is thus that there must be an intermediate distance at which there is neither repulsion nor attraction.

# Chemical Bonding

- Chemical compounds are formed by the joining of two or more atoms.
- A stable compound occurs when the total energy of the combination has **lower energy** than the separated atoms.
- The bound state implies a net attractive force between the atoms ... a chemical bond.
- The two extreme cases of chemical bonds are:
- **Covalent** bond: bond in which one or more pairs of electrons are **shared** by two atoms.
- **Ionic** bond: bond in which one or more electrons from one atom are **removed and attached** to another atom, resulting in positive and negative ions which attract each other.
- Other types of bonds include metallic bonds and hydrogen bonding. The attractive forces between molecules in a liquid can be characterized as van der Waals bonds.

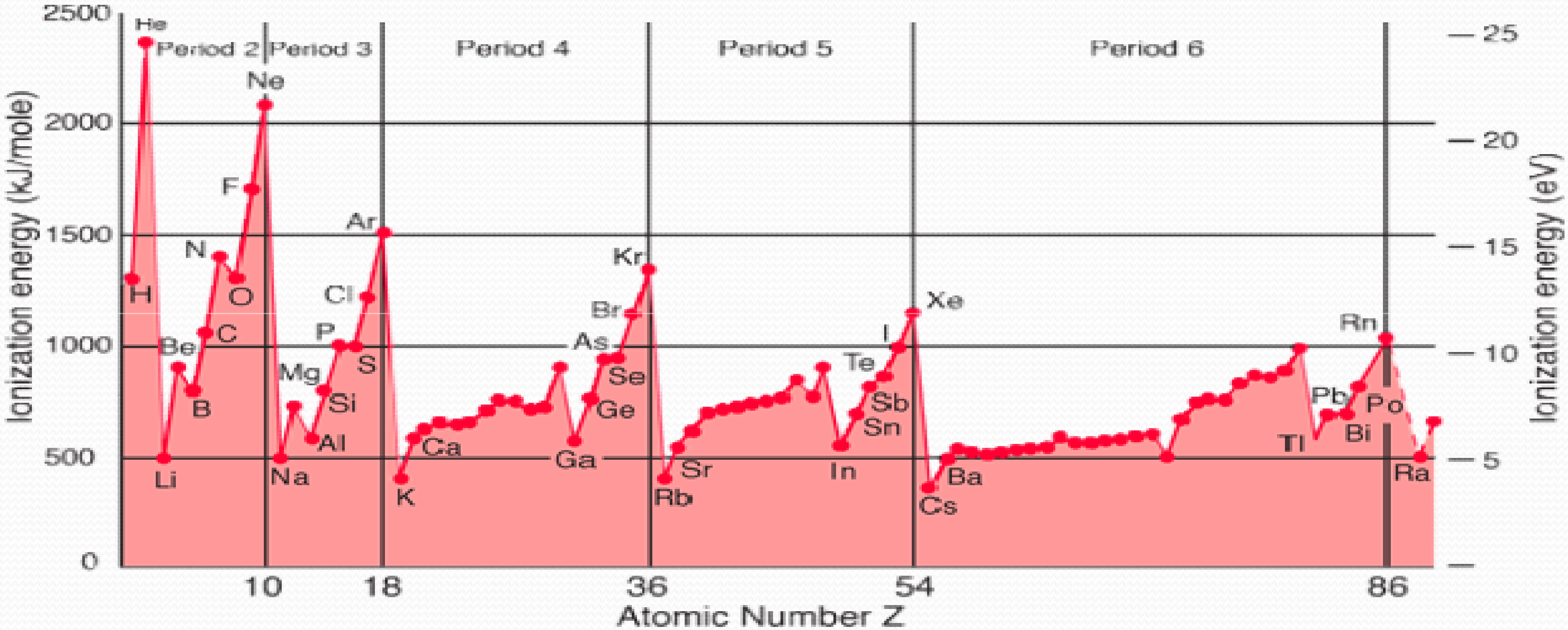
# Ionization energy

- we can describe ionization energy as the minimum energy that an electron in a gaseous atom or ion has to absorb to come out of the influence of the nucleus. It is also sometimes referred to as ionization potential and is usually an endothermic process.
- **The ionization energy** or ionization potential is the energy necessary to **remove** an electron from the neutral atom.

- ionization energy gives us an idea of the **reactivity** of chemical compounds.
- used to determine the **strength** of chemical bonds.
- units of electronvolts or kJ/mol.
- Depending on the ionization of molecules which often leads to changes in molecular geometry, ionization energy can be either adiabatic ionization energy or vertical ionization energy.

# Ionization energy

- It is a minimum for the alkali metals which have a single electron outside a closed shell.
- It generally increases across a row on the periodic maximum for the noble gases which have closed shells.
- For example, sodium requires only 496 kJ/mol or 5.14 eV/atom to ionize it while neon, the noble gas immediately preceding it in the periodic table, requires 2081 kJ/mol or 21.56 eV/atom.
- The ionization energy can be thought of as a kind of counter property to **electronegativity** in the sense that a **low** ionization energy implies that an element readily **<u>gives</u>** electrons to a reaction, while a **high** electronegativity implies that an element strongly seeks to **<u>take</u>** electrons in a reaction.



Ionization energy (kJ/mole)
Ionization energy (eV)
Atomic Number Z
Period 2
Period 3
Period 4
Period 5
Period 6
2500
2000
1500
1000
500
0
25
20
15
10
5
10
18
36
54
86
He
H
Li
Be
B
C
N
O
F
Ne
Na
Mg
Al
Si
P
S
Cl
Ar
K
Ca
Ga
Ge
As
Se
Br
Kr
Rb
Sr
In
Sn
Sb
Te
I
Xe
Cs
Ba
Tl
Pb
Bi
Po
Rn
Ra

# Factors that govern the attraction forces.

- If the nucleus is positively charged then the electrons are strongly attracted to it.
- If an electron lies near or close to the nucleus then the attraction will be greater than the one when the electron is further away.
- If there are more electrons between the outer level and the nucleus the attraction forces are less.
- When there are two electrons in the same orbital they experience some form of repulsion. Now, this creates disturbances in the attraction of the nucleus. In essence, ionization energy will be less in paired electrons as they can be removed easily.

# How to calculate?

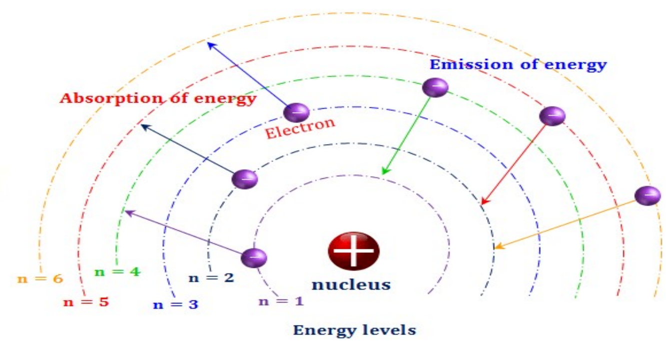
- Energy of an electron in 'n'th orbit is calculated by Bohr model of an atom as –

$$E_n = -\frac{2\pi^2 m e^4}{(4\pi\varepsilon^o)^2 h^2} \times \frac{Z^2}{n^2} = R \times \frac{Z^2}{n^2} J/atom = -13.6 \times \frac{Z^2}{n^2} eV/atom = -2.18 \times 10^{-18} \times \frac{Z^2}{n^2} J/atom$$

Ionization energy for the removal of an electron from a neutral atom can be calculated, by substituting, the orbit number of the electron before transition as '$n_1$' and orbit number of the electron after transition as '∞'( infinity) as '$n_2$' in Bohr's energy equation.

$$En_1 = -R \times \frac{Z^2}{n^2} \qquad En_2 = -R \times \frac{Z^2}{\infty^2}$$

$$\Delta E = En_2 - En_1 = R \times z^2 \left(\frac{1}{n^2} - \frac{1}{\infty^2}\right) = ionization\ Energy$$




Since,

$\frac{1}{\infty^2}$

is almost zero, it can be neglected.

$$Ionization\ Energy = \Delta E = En_2 - En_1 = R \times z^2 \left(\frac{1}{n^2}\right) = 2.18 \times 10^{-18} \times z^2 \left(\frac{1}{n^2}\right) J$$

# Problem

- 1. Electromagnetic radiation of wavelength 242 nm is just sufficient to ionize the sodium atom. Calculate the ionization energy of sodium in kJ $mol^{-1}$.

# Electron Affinity

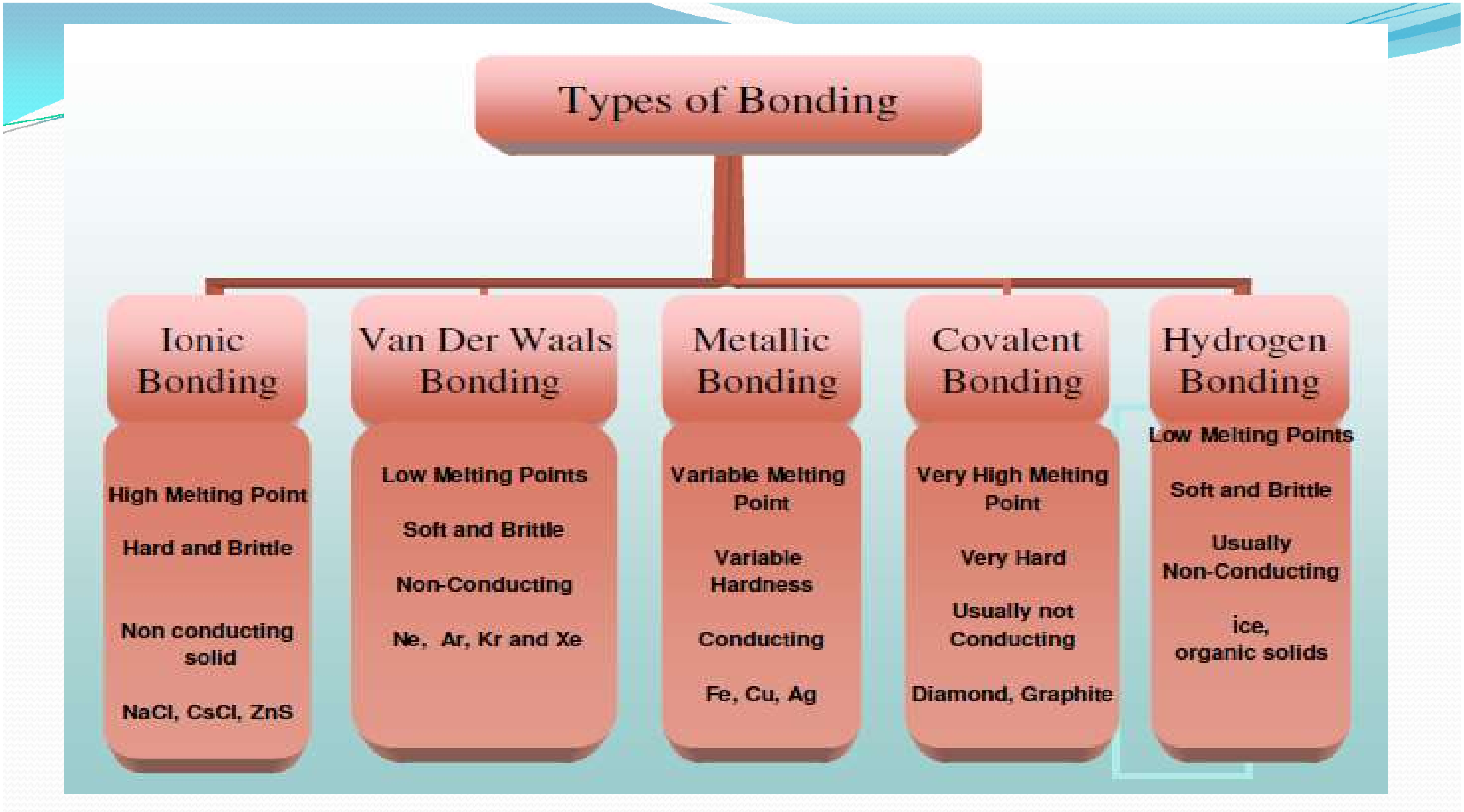
- The electron affinity is a measure of the energy change when an electron is **added** to a neutral atom to form a negative ion.
- For example, when a neutral chlorine atom in the gaseous form picks up an electron to form a Cl- ion, it releases an energy of 349 kJ/mol or 3.6 eV/atom.
- It is said to have an electron affinity of -349 kJ/mol and this large number indicates that it forms a stable negative ion.
- Small numbers indicate that a less stable negative ion is formed.
- Groups VIA and VIIA in the periodic table have the largest electron affinities.

# Electronegativity

- Electronegativity is a measure of the ability of an atom in a molecule to **draw bonding electrons** to itself.
- The most commonly used scale of electronegativity is that developed by Linus Pauling in which the value 4.0 is assigned to fluorine, the most electronegative element. (Oxygen 3.44)
- Lithium, at the other end of the same period on the periodic table, is assigned a value of 1.
- Electronegativity generally increases from left to right on the periodic table and decreases from top to bottom.
- Metals are the least electronegative of the elements.
- The Pauling electronegativities for the elements are often included as a part of the chart of the elements.

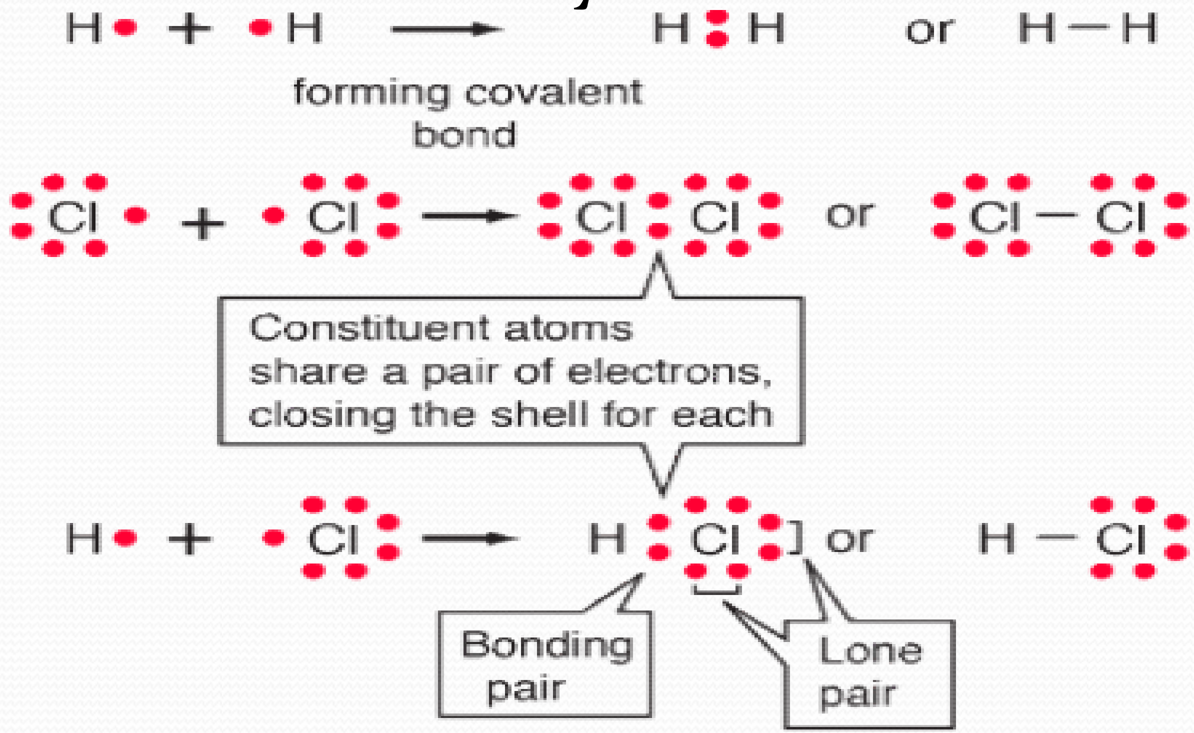
**Table 2.2** Comparison of electronegativity with the sum of ionization energy and electron affinity. All values are for 25°C

| | $E_I$ | $E_A$ | Sum | $e_N$ |
|---|---|---|---|---|
| F | 2.8 | 0.6 | 3.4 | 4.0 |
| Cl | 2.1 | 0.6 | 2.7 | 3.0 |
| Br | 1.9 | 0.5 | 2.4 | 2.8 |
| I | 1.7 | 0.5 | 2.2 | 2.5 |
| H | 2.2 | 0.1 | 2.3 | 2.1 |
| Li | 0.9 | 0.1 | 1.0 | 1.0 |
| Na | 0.8 | 0.1 | 0.9 | 0.9 |
| K | 0.7 | 0 | 0.7 | 0.8 |
| Rb | 0.7 | 0 | 0.7 | 0.8 |
| Cs | 0.6 | 0 | 0.6 | 0.7 |


Types of Bonding
Ionic Bonding
High Melting Point
Hard and Brittle
Non conducting solid
NaCl, CsCl, ZnS
Van Der Waals Bonding
Low Melting Points
Soft and Brittle
Non-Conducting
Ne, Ar, Kr and Xe
Metallic Bonding
Variable Melting Point
Variable Hardness
Conducting
Fe, Cu, Ag
Covalent Bonding
Very High Melting Point
Very Hard
Usually not Conducting
Diamond, Graphite
Hydrogen Bonding
Low Melting Points
Soft and Brittle
Usually Non-Conducting
ice, organic solids

**Covalent Bonds**

Covalent chemical bonds involve the sharing of a pair of valence electrons by two atoms

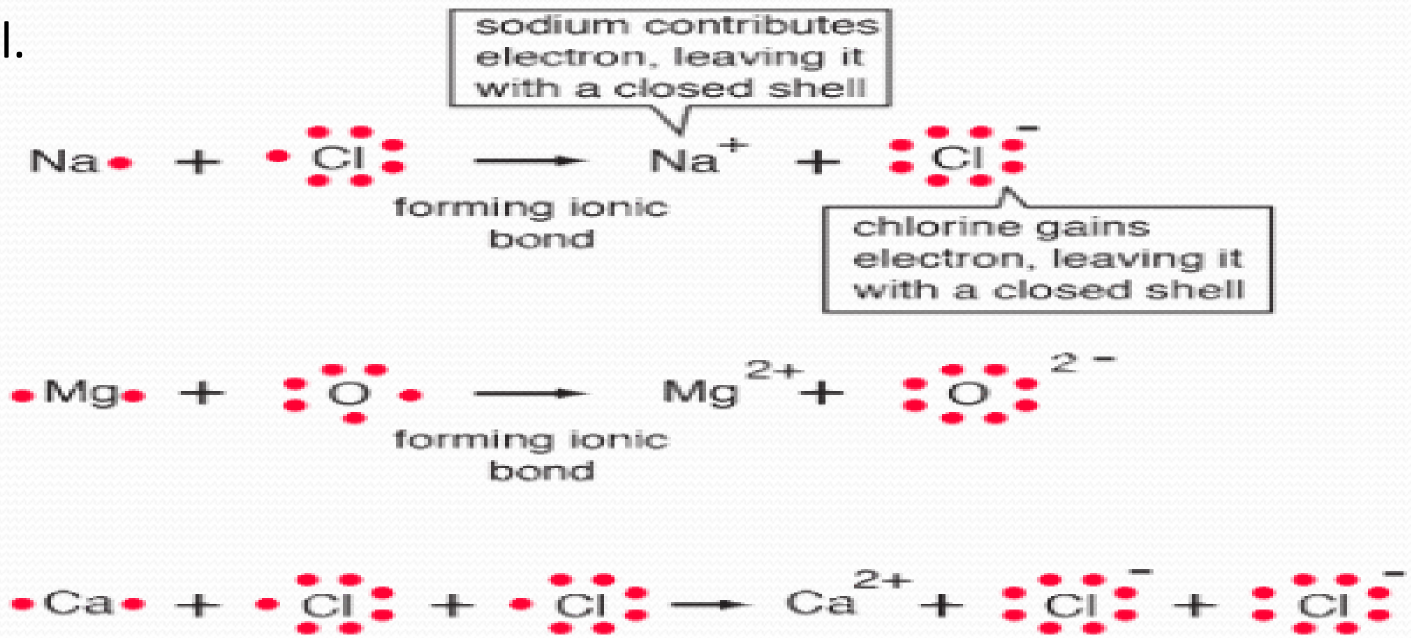


# Polar Covalent Bonds

- Covalent bonds in which the sharing of the electron pair is unequal, with the electrons spending more time around the more nonmetallic atom, are called polar covalent bonds.
- In such a bond there is a charge separation with one atom being slightly more positive and the other more negative, i.e., the bond will produce a dipole moment.
- The ability of an atom to attract electrons in the presence of another atom is a measurable property called electronegativity.

# Ionic Bonds

- In chemical bonds, atoms can either transfer or share their valence electrons. In the extreme case where one or more atoms lose electrons and other atoms gain them in order to produce a noble gas electron configuration, the bond is called an ionic bond.
- Typical of ionic bonds are those in the alkali halides such as sodium chloride, NaCl.



# Metallic Bonds

- The properties of metals suggest that their atoms possess strong bonds ,conduction of heat and electricity suggest that electrons can move freely in all directions in a metal.
- They are neither ionic nor covalent since the participating electrons are not localized on the atoms.

https://app.playmada.com/

# Secondary Bonds

## 1.VAN DER WAALS BONDING

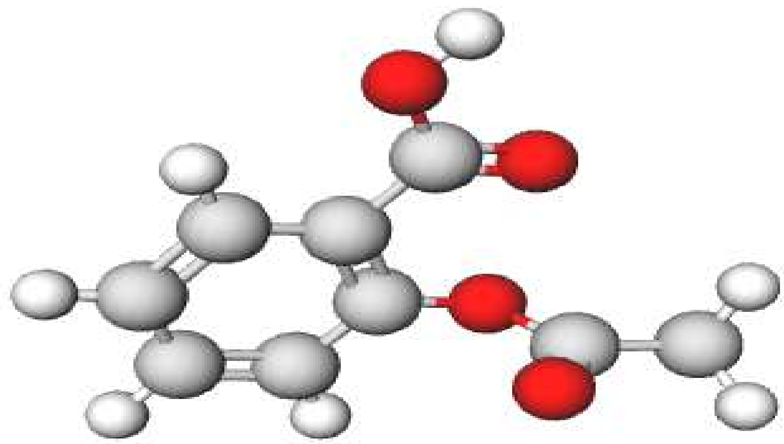
- It is a weak bond, with a typical strength of 0.2 eV/atom.
- It occurs between neutral atoms and molecules.
- The explanation of these weak forces of attraction is that there are **natural fluctuation in the electron density** of all molecules and these cause small **temporary dipoles** within the molecules.
- It is these temporary dipoles that attract one molecule to another. They are called van der Waals' forces.
- The bigger a molecule is, the easier it is to polarise (to form a dipole), and so the van der Waal's forces get stronger, so bigger molecules exist as liquids or solids rather than gases.

- The **shape of a molecule influences its ability to form** temporary dipoles. Long thin molecules can pack closer to each other than molecules that are more spherical.
- The bigger the 'surface area' of a molecule, the greater the van der Waal's forces will be and the higher the melting and boiling points of the compound will be.
- Van der Waal's forces are of the order of 1% of the strength of a covalent bond.
- The dipoles can be formed as a result of unbalanced distribution of electrons in asymettrical molecules. This is caused by the instantaneous location of a few more electrons on one side of the nucleus than on the other.
- Therefore atoms or molecules containing dipoles are attracted to each other by electrostatic forces.
- These forces are due to the electrostatic attraction between the nucleus of one atom and the electrons of the other.
- Van der waals interaction occurs generally between atoms which have noble gas configuration.

# 2.Hydrogen bond

- The other type of secondary bond that we need to consider here is the hydrogen bond, and its importance in biological structures could hardly be exaggerated.
- It frequently arises through the interaction of hydrogen and oxygen atoms, as in the configuration NH-OC.
- The electronegativity of the nitrogen atom is so large compared with that of the hydrogen that the hydrogen atom’s lone electron tends to spend most of its time between the two atomic nuclei.
- This means that the (proton) nucleus of the hydrogen atom is not so well screened by the electron as it would be in the isolated neutral version of that atom.
- The net result is that the hydrogen atom develops a positive pole.
- Just the reverse happens in the case of the oxygen atom, the electronegativity of which is greater than that of the carbon atom, so it, in turn, functions as a negative pole.

- The hydrogen bond is not very strong, but it is quite directional, in that the nucleus of the hydrogen atom is exposed only within a limited solid angle.
- Moreover, the small size of the hydrogen atom limits the number of other atoms that can approach it sufficiently closely.
- The hydrogen bond is thus also quite saturated.
- These forces dictate the spatial arrangement of the atoms in a molecule.
- If we wish to make calculations on the relative stabilities of various candidate conformations, therefore, we will need a quantitative description of these forces.



- How many hydrogens in figure  can form hydrogen bonds?

Only one, the one at the very top which is attached to the highly electrongative oxygen atom (red), all the others are attached to carbon and can not hydrogen bond.

# H-Bonds and Water

- H-bonding occurs in water. In the liquid state they are rapidly being formed and broken as the mobile particles move over each other.
- Note in figure that there are two type of O-H bonds, the intramolecular O-H bond within a molecule (bond length = 1.01Å) and the intermolecular bond between atoms (bond length = 1.75Å).
- The closeness of the bond length indicates that the intramolecular bond is very strong, and of comparable magnitude to the intramolecular one.
- On a macroscopic scale this is obvious to anyone who falls on ice (frozen water) and quickly realizes how hard it is, and how strong the bonds are that hold the water molecules to each other.

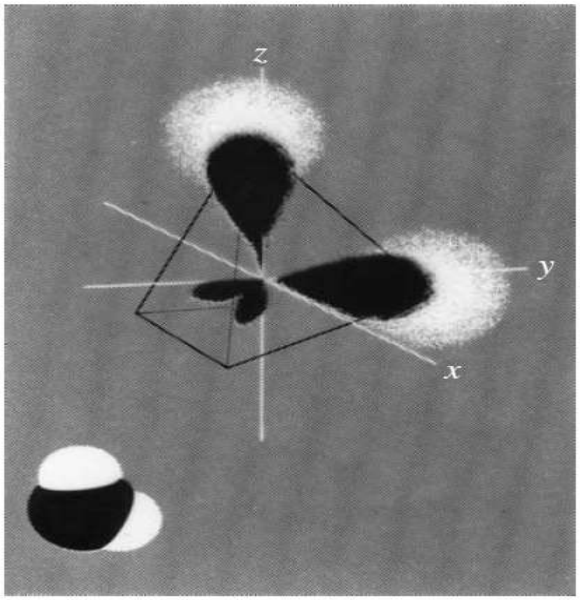
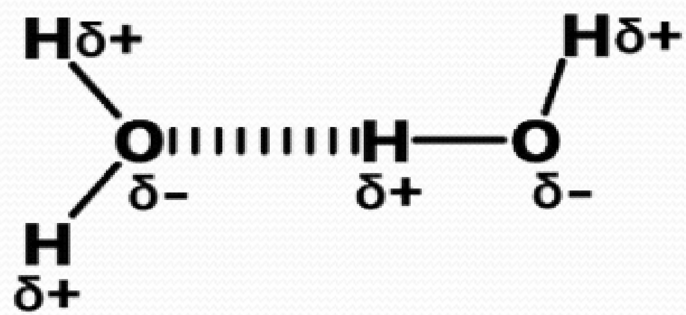
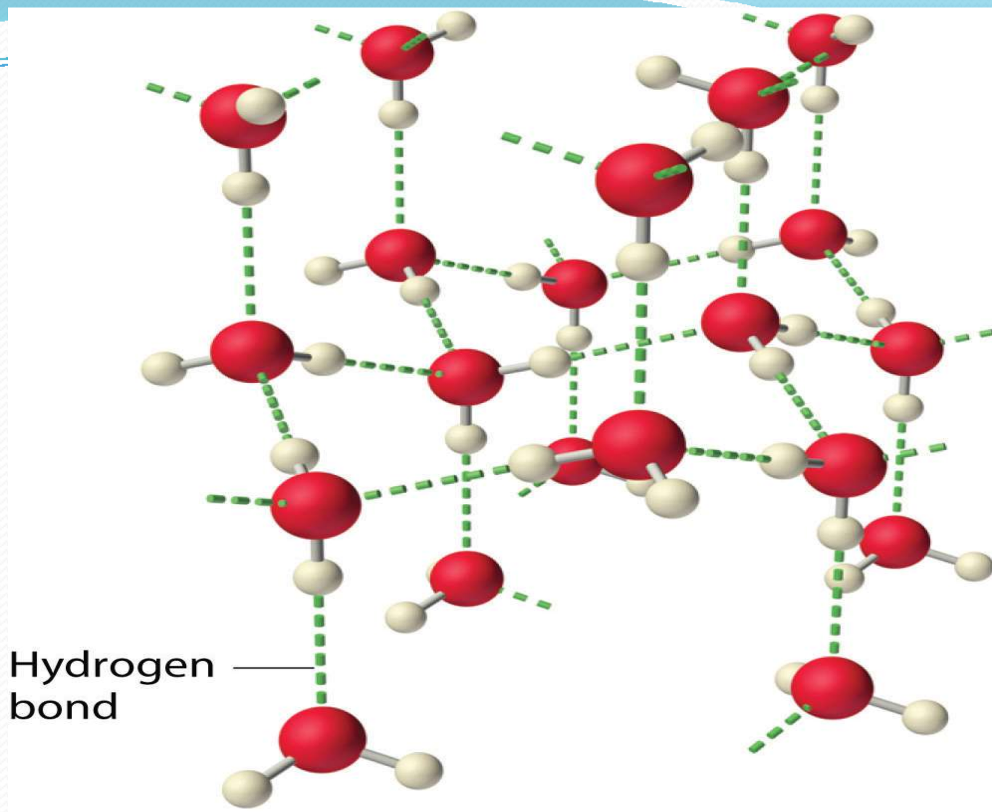



**Figure 2.8** Hybridization in the water molecule, $H_2O$



- Water also has two lone pairs and two H atoms attached to the highly electronegative oxygen.
- This means each water molecule can participate in up to 4 bonds (two where it is the h-bond acceptor, and two where it is the h-bond donor).
- One interesting consequence of this is that water forms a 3D crystalline structure that is sort of based on a distorted tetrahedron.  That is, the oxygen is $sp^3$ hybridized with a tetrahedral electronic geometry, having two bonding orbitals and two lone pairs.
- All of these are involved with hydrogen bonds. The lone pairs are functioning as H-bond acceptors, and the hydrogen on the bonding orbitals are functioning as h-bond donors.
- So each oxygen is attached to 4 hydrogens, two are 1.01Å covalent bonds and two are 1.75 Å hydrogen bonds, and this results in a structure like figure, which has lots of void space, and the consequence that ice is less dense than liquid water and floats.



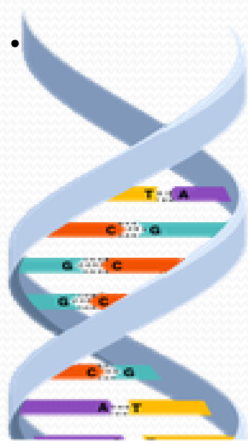

- The fact that ice floats has great ramifications for life. The ice, with it's void space, acts as an insulator. If the ice sunk to the bottom, lakes would completely freeze and aquatic life like fish would not be able to survive the winters.

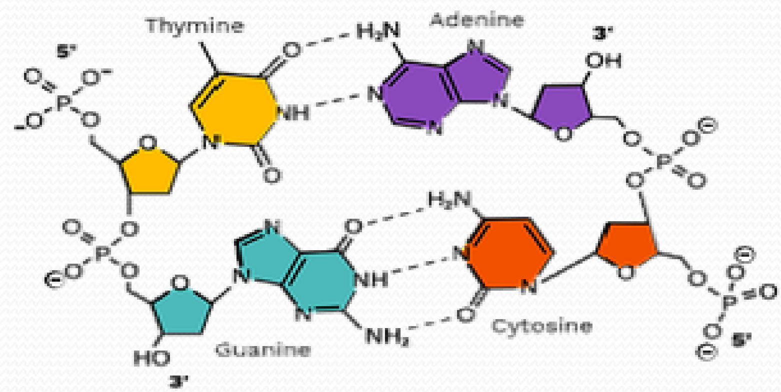
# H-bonding and Boiling Points

- Boiling points are an indicator of intermolecular forces,
- water has a higher boiling point than HF, yet fluorine is more electronegative than oxygen, it is also smaller, and so you would expect the HF hydrogen bond to be stronger than the OH hydrogen bond.
- In fact, this is all try, and what this argument does not take into account is the number of hydrogen bonds. That is, the cohesive forces that hold the liquid together are not just the strength of the bonds, but also the number of bonds.
- Each HF molecule has one H, and 3 lone pairs on the fluorine. So in a macroscopic system like a mole of HF, each HF would on average be involved with 2 bonds, one involving the hydrogen (hydrogen donor), and one involving a fluorine lone pair (hydrogen acceptor), and simple speaking, there are not enough hydrogens to use up all the lone pairs, (two of the fluorine's lone pairs are not involved in H bonds.)
- In the case of water, the number of lone pairs equals the number of hydrogens, and so each water molecule can on the average, be involved with 4 hydrogen bonds.
- So even though the individual hydrogen bonds in water may be weaker than in HF, there are more of them, making the boiling point of water higher than HF.

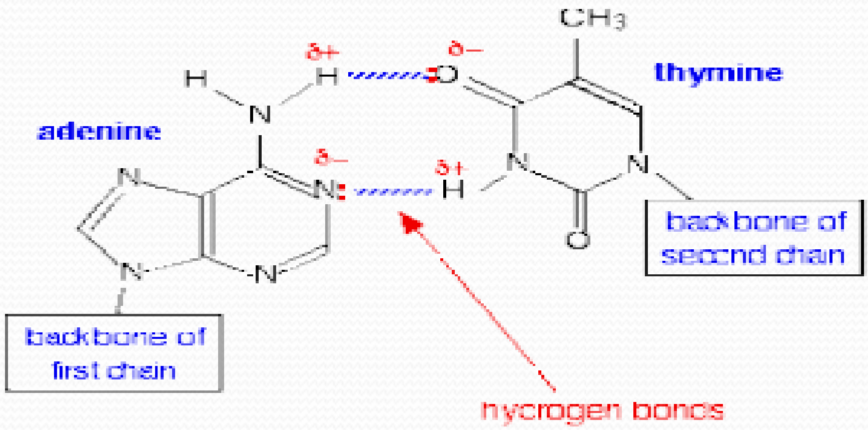
# Hydrogen Bonding and Biology

- We have all heard of DNA, which consists of nucleotide strands that join together to form the infamous double helix.
- DNA helix showing the base pairs adenine/thymine (A-T) an Guanine/Cytosine. (G-C), which form the double helix through the base pair interactions that are made through hydrogen bonding .

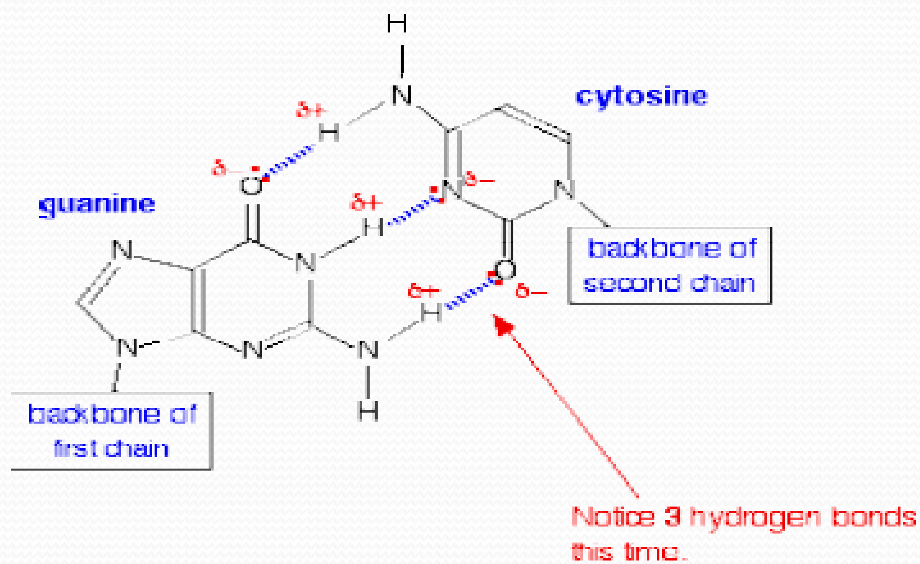




## The adenine-thymine base pair held together by two hydrogen bonds(left) and the guanine-cytosine base pair held together by three hydrogen bonds (right).

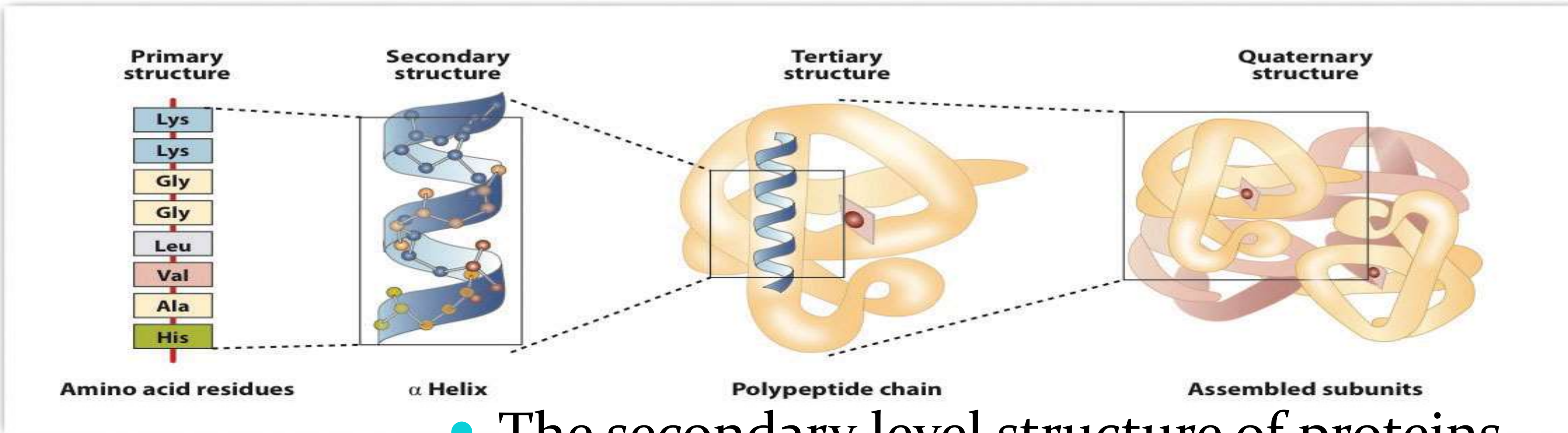



**GΞC**

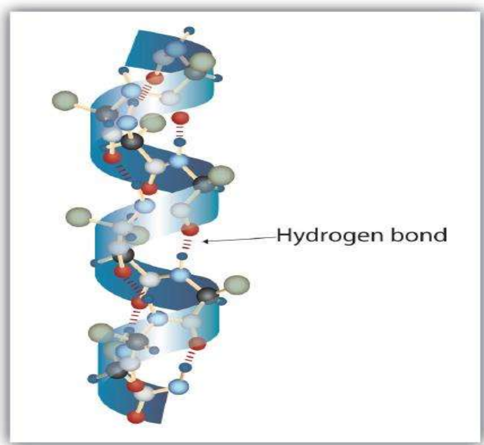


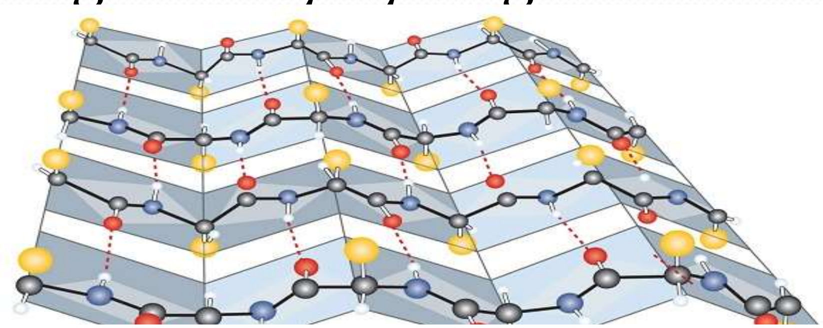

**A=T**

## Proteins are based on amino acids and have 4 structural levels




- The secondary level structure of proteins form alpha helixes and beta-pleated sheets, which are held together by hydrogen bonds.



# Hydrogen bonds contribution

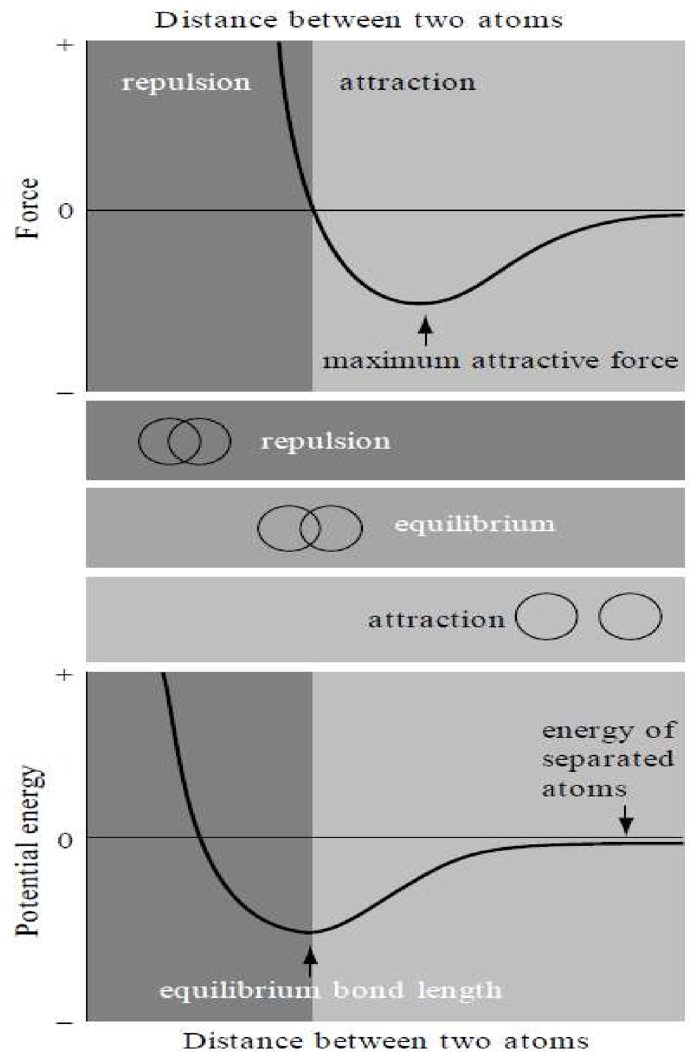
- Highly specific
- 1-2 kcal/mol per hydrogen bond
- Not very strong

# Interatomic Potentials for Strong Bonds

- Ludwig Seeber in 1824

- Graphical representation of the relationship between the force between two atoms and the distance between them

$$\mathcal{F}(r) = -\frac{\partial \mathscr{E}(r)}{\partial r}$$

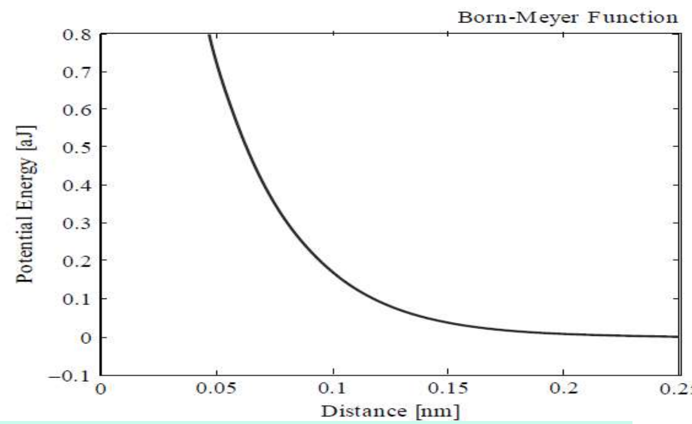


# Born–Mayer function

- a *component of the interatomic interaction was made by* Max Born and Joseph Mayer in 1932
- Using the techniques of quantum mechanics

closed-shell repulsion potential has the form

$$\mathscr{E}(r) = A \exp\left(\frac{2r_s - r}{\rho}\right)$$




The Born–Mayer function is analytically exact as a description of the repulsive interaction between two atoms, each of which exclusively consists of closed electron shells

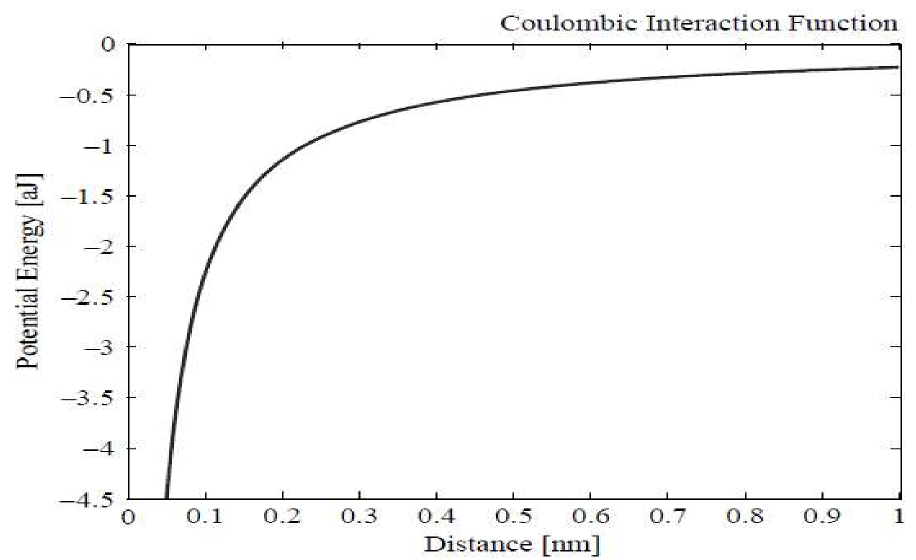
# Coulomb interaction

- When two charged atoms, that is to say two ions, interact, a Coulomb contribution to the interatomic potential will always be present.

$$\mathscr{E}(r) = \frac{1}{4\pi\epsilon_0}\frac{\tau_1\tau_2 q_e^2}{r}$$

$$\mathscr{E}(r) = \frac{-1}{4\pi\epsilon_0}\frac{q_e^2}{r}\exp\left(\frac{-r}{\zeta}\right)$$

Coulomb interaction modulated by an exponentially decreasing factor, and the negative sign reflects the fact that the potential is always attractive.



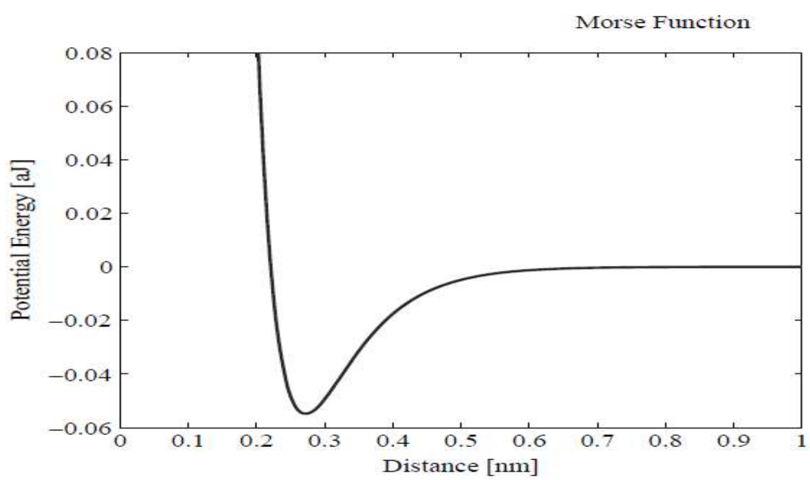
# Morse potential function

- there are terms which arise from the above-mentioned electron interference and scattering effects.
- In calculations on the properties of metals
- the Morse potential has both repulsive and attractive terms

$$\mathscr{E}(r) = \mathscr{E}_{\text{Morse}}\{\exp[-2\alpha(r-r_0)] - 2\exp[-\alpha(r-r_0)]\}$$

the potential really does have its minimum value at *ro*

$$\left.\frac{\partial \mathscr{E}(r)}{\partial r}\right|_{r=r_0} = 0$$



# Interatomic Potentials for Weak Bonds

- Van der Waals interaction

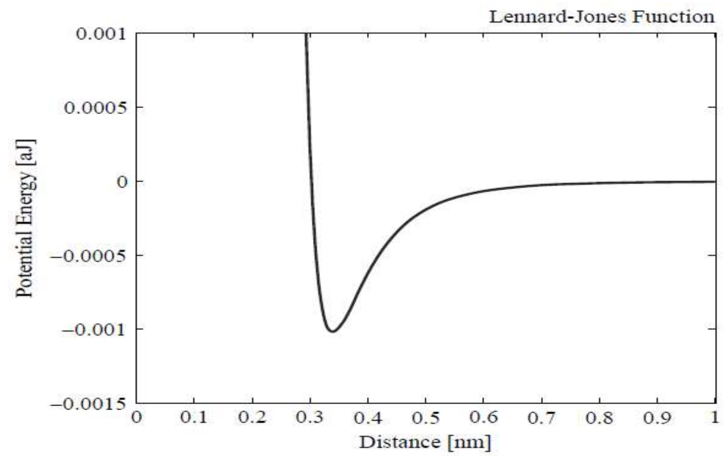
$$\mathscr{E}(r) = -\beta / r^6$$

- Buckingham potential function

$$\mathscr{E}(r) = A \exp(-r/\rho) - \beta / r^6$$

# Lennard-Jones potential function

$$\mathcal{E}(r) = \varepsilon\{(r_0/r)^{12} - 2(r_0/r)^6\}$$




- ➢Lennard-Jones potential function extremely popular in the computer simulation of all manner of atomic and molecular systems
- ➢the energy minimum occurs at a distance $r_o$, *and that* the depth of the potential well is simply -ε

$$\left.\frac{\partial \mathcal{E}(r)}{\partial r}\right|_{r=r_0} = 0$$

- ➢the interatomic potential, by convention, always goes to zero as *r tends to infinity.*

$$\mathcal{E}(r)|_{r\to\infty} = 0$$

**Table 3.2** The Lennard-Jones distance parameter for the non-bonded interactions between pairings of atoms commonly encountered in the biological domain

| $r_0$ (nm) | C | N | O | H | S |
|---|---|---|---|---|---|
| C | 0.34 | 0.33 | 0.33 | 0.28 | 0.35 |
| N | | 0.31 | 0.31 | 0.28 | 0.33 |
| O | | | 0.30 | 0.27 | 0.33 |
| H | | | | 0.24 | 0.30 |
| S | | | | | 0.30 |

Distance parameter

**Table 3.3** The Lennard-Jones energy parameter for the non-bonded interactions between pairings of atoms commonly encountered in the biological domain

Energy parameter

| $\varepsilon 10^{-3}$ aJ | C | N | O | H | S |
|---|---|---|---|---|---|
| C | 1.02 | 1.58 | 1.41 | 0.80 | 2.18 |
| N | | 2.50 | 2.22 | 1.22 | 3.40 |
| O | | | 2.00 | 1.17 | 3.05 |
| H | | | | 0.78 | 1.72 |
| S | | | | | 2.10 |

# Hydrogen-bond potential functions

by Frank Stillinger and one that has been fairly popular is the following

$$\mathscr{E}(r) = \mathscr{E}_{LJ}(r) + \mathscr{E}_{ionic}(r) + \mathscr{E}_{s}(r)$$

in which $\mathscr{E}_{LJ}(r)$ is the Lennard-Jones potential (with parameters chosen to correspond to the non-bonding situation), $\mathscr{E}_{ionic}(r)$ is the Coulombic interaction given by Equation (3.3), and $\mathscr{E}_{s}(r)$ is known as a switching function. The effect of the latter is to make the overall $\mathscr{E}(r)$ go smoothly to zero for distances greater than a critical value, which is usually taken to be 0.285 nm.

# Non-central Forces

- Complexities arise when one takes non-central components of the interatomic force into account.
- an empirical approach, using approximations where necessary.
- an expression for the total energy of a molecule that is now widely used in computer simulations of biological molecules has the following form

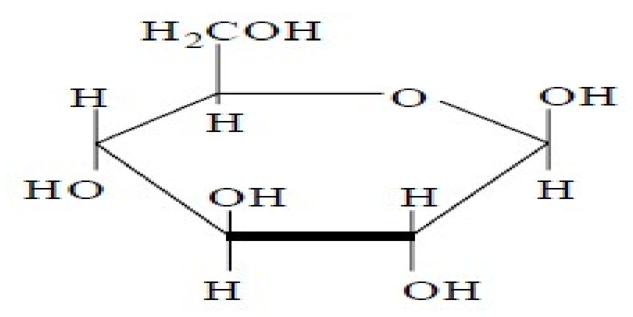
$$
\begin{aligned}
\mathscr{E}_{\text{total}} = & \sum_{b} [k_2(b-b_0)^2 + k_3(b-b_0)^3 + k_4(b-b_0)^4] \\
& + \sum_{\theta} [k_2(\theta-\theta_0)^2 + k_3(\theta-\theta_0)^3 + k_4(\theta-\theta_0)^4 \\
& + \sum_{\phi} [k_1(1-\cos\phi) + k_2(1-\cos 2\phi) + k_3(1-\cos 3\phi) \\
& + \sum_{\chi} k_2\chi^2 + \sum_{\mathrm{b,b'}} k(b-b_0)(b'-b'_0) \\
& + \sum_{b,\theta} k(b-b_0)(\theta-\theta_0) + \sum_{b,\phi} (b-b_0)[k_1\cos\phi + k_2\cos 2\phi + k_3\cos 3\phi] \\
& + \sum_{\theta,\phi} (\theta-\theta_0)[k_1\cos\phi + k_2\cos 2\phi + k_3\cos 3\phi] \\
& + \sum_{\theta,\theta'} k(\theta-\theta_0)(\theta'-\theta'_0) + \sum_{\theta,\theta',\phi} k(\theta-\theta_0)(\theta'-\theta'_0)\cos\phi \\
& + \sum_{i,j} \frac{q_j q_j}{r_{ij}} + \sum_{i,j} \varepsilon_{ij}\left[2\left(\frac{r_{ij}^0}{r_{ij}}\right)^9 - 3\left(\frac{r_{ij}^0}{r_{ij}}\right)^6\right]
\end{aligned}
$$

- Such a potential can be regarded as deriving from two types of interaction: the valence terms and the non-bonded terms, the latter referring to those components that do not stem from covalent bonds.
- The valence terms themselves also come from different categories of interaction, namely what are called the diagonal components and the off-diagonal cross-coupling components.
- In the valence terms, there are the internal coordinates of bond length, *b, angle,* $\theta$, torsion angle, $\phi$, and out-of-plane angle, $\chi$.
- The cross-coupling terms include the various possible combinations of two and three internal co-ordinates, and they are found to be important if good agreement with experimental vibration spectra is at a premium.
- This is generally the case, because large biological molecules are not static at body temperature; their dynamic properties are of great significance for metabolic function.
- The non-bonded terms include a Lennard-Jones type function that has a ninth rather than a 12$^{th}$ power in its repulsive part; the repulsion is thus somewhat softer.

# Bond Energies

- $5 \times 10^{22}$ atoms per gram of tissue
- three bonds per atom
- an adult human would comprise about $4 \times 10^{27}$ bonds.
- A typical covalent bond energy is approximately 0.6aJ (i.e. $0.6 \times 10^{-18}$ J)
- Total energy of all the bonds in the body is about $2.4\ \ 10^{9} \times$ J or $0.6 \times 10^{9}$ calories
- 250 times the normal daily intake of 2400 kilocalories

| Bond | Energy (aJ) | Distance (nm) |
|---|---|---|
| H–H | 0.72 | 0.074 |
| H–C | 0.69 | 0.109 |
| H–N | 0.65 | 0.100 |
| H–O | 0.77 | 0.096 |
| C–C | 0.58 | 0.154 |
| C=C | 1.02 | 0.135 |
| C≡C | 1.39 | 0.121 |
| C–N | 0.49 | 0.147 |
| C=N | 1.02 | 0.129 |
| C≡N | 1.48 | 0.115 |
| C–O | 0.58 | 0.143 |
| C=O | 1.23 | 0.122 |
| N–N | 0.26 | 0.148 |
| N=N | 0.69 | 0.124 |
| N≡N | 1.57 | 0.100 |
| N–O | 0.35 | 0.144 |
| N=O | 0.80 | 0.120 |
| O–O | 0.35 | 0.149 |
| O=O | 0.82 | 0.128 |



| Number | Type | Energy (aJ) |
|---|---|---|
| 5 | C–C | 2.90 |
| 7 | C–O | 4.06 |
| 7 | C–H | 4.83 |
| 5 | O–H | 3.85 |
| Total | = | 15.64 |

equation for the respiration process

$$C_6H_{12}O_6 \quad + \quad 6O_2 \quad \rightarrow \quad 6CO_2 \quad + \quad 6H_2O \quad + \quad 3.44\ \text{aJ}$$

$-15.64\ \text{aJ}$

The well-known polymer polyisoprene occurs with two different arrangements of its side-chains. These are known as the *cis- and trans-isomers. They* are commonly known as natural rubber and gutta-percha, respectively. The noteworthy thing is that whereas natural rubber is very flexible, gutta-percha is stiff and hard. (Interestingly, both these isomers were used in the golf balls of earlier decades, the natural rubber providing the elastic core and the guttapercha being used for the stiffer outer cover.)

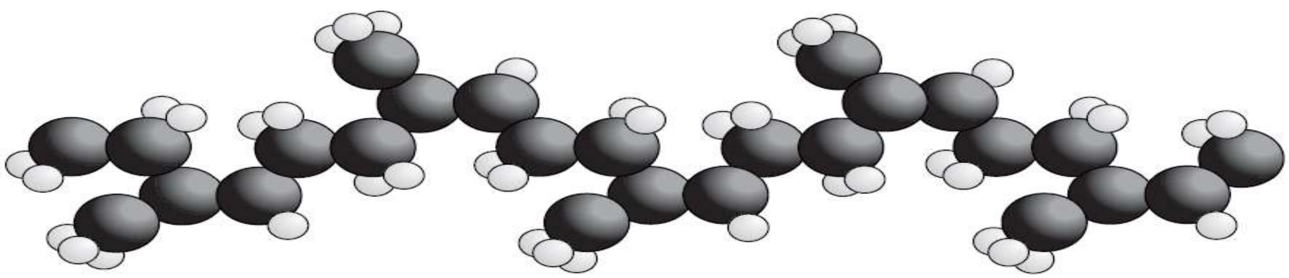


**Figure 3.8** The naturally kinked molecular configuration of *cis*-polyisoprene gives it a high degree of elasticity

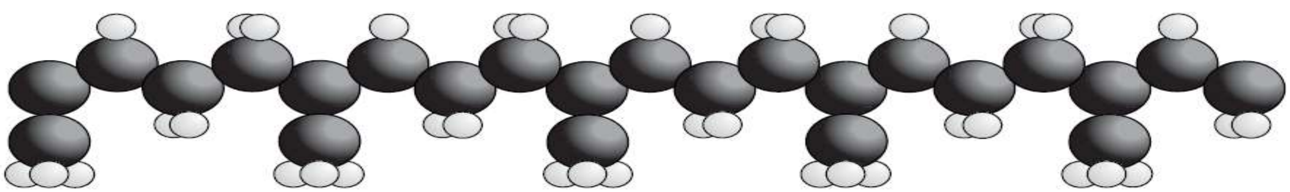


**Figure 3.9** Molecules of *trans*-polyisoprene do not have the kinked structure of their *cis* counterparts and they are far more rigid

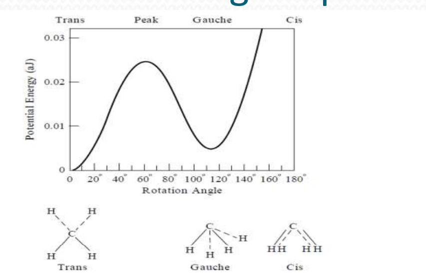



**Figure 3.10** The experimental rotational energy function for twisting about one of the C–C bonds in polyethylene

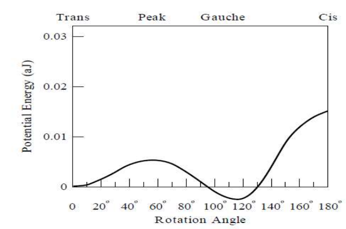



**Figure 3.11** The calculated rotational energy function for twisting about one of the bonds in polyethylene using Tables 3.2 and 3.3

# Spring Constants

- Until now, we have been considering the forces that act at the atomic and molecular level. In many biophysical problems, however, we need to know something about the forces that act on a more macroscopic level.
- These forces are ultimately due to those atomic and molecular level forces, of course, but we need to be able to approximate the interactions that occur at the grosser level, without having to think about their microscopic roots.
- These were the issues that occupied materials scientists of earlier centuries, in fact, and they made commendable progress despite the primitive nature of science in those days. One of the early giants in this type of endeavour **was Robert Hooke, who established that there is a linear relationship between the stress applied to a piece of material and the resulting strain.** This wisdom is embodied in what came to be known as Hooke's Law, and he showed that the simple proportionality holds as long as the degree of strain is small.
- it breaks down when the material is strained beyond what is known as the elastic limit. The science of those early days tended to be written up in Latin, and Hooke's result is immortalized in the phrase *ut tensio sic vis (the elongation is as the force).*
- When dealing with such multi-atom structures as the components of muscle fibres, for example, it is customary to invoke Hookian behaviour, and to assume that the macroscopic restoring force is linearly related to the strain, by an expression of the type
- ***F _ restoring =- κΔ x***

the negative sign arising because the force acts in the direction counter to the extension *x.*

*The coefficient κ is known as a spring constant*

# Internal Energy

- *thermodynamic potential energies are calculated relative to the minimum in the potential well.*
- *Internal energy is defined by the* following equation
- $E_{int} = E_{pot} + E_{kin}$
- in which $E_{pot}$ is the potential energy, and $E_{kin}$ is the kinetic energy.
- considering a system of just two atoms. When they are at their equilibrium separation, and stationary,
- there is neither potential nor kinetic energy, so the internal energy is simply zero.

The kinetic energy arises from the motions of the individual atoms, and the formal definition of this parameter for a system comprising *N atoms is simply*

$$\mathscr{E}_{\text{kin}} = \sum_{i=1}^{N} \frac{m}{2} v_i^2$$

If the system is a three-dimensional one, which is to say that the positions of all the atoms require three different variables to describe them completely, the relationship between the velocities and the temperature is as given by the following equation

$$\mathscr{E}_{\text{kin}} = \left(\frac{3}{2}\right) N k_B T$$

in which $k_B$ *is the Boltzmann constant. The latter has the value 1.38054 x* $10^{-23}$ $JK^{-1}$.

When a situation of equipartition prevails, there will be equality between the potential energy and the kinetic energy. However, this can be the case only if the particles interact through a strictly harmonic potential (that is to say that the inter-particle interaction varies as the square of the separation distance).

- In general, the breaking of interatomic bonds requires more energy than the mere redistribution of these.
- The stronger types of bond obviously require more energy expenditure if they are to be ruptured.
- If the bond-breaking is to be accomplished thermally, the stronger bonds will need warming to higher temperatures for a given rate of bond-breaking.
- In fact, the situation is more complicated because many atoms can act cooperatively to break a bond that would be beyond the disruptive capability of a single moving atom.
- This is seen in enzymes, in which the energy of many atoms is concentrated at a particular place.

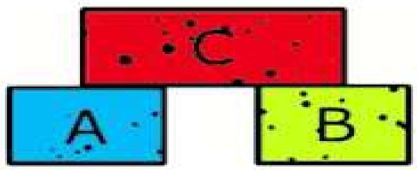
- One might think that covalent bonds, which are much stronger than hydrogen bonds, would not be vulnerable to breaking in this manner, but the work of Alexander Fleming on bacteria suggested that this can indeed happen.
- One day, when he was suffering from a cold, a few drops of nasal mucus fell from his nose onto a culture of bacteria. Ever inquisitive, he decided not to throw the culture away, and a few days later he was surprised to find that the bacteria had been killed in the very area where the mucus had fallen. These bacteria were protected by an outer shell, which comprised many covalent bonds.
- Something in the mucus had apparently been capable of breaking these bonds, and the enzyme in question is now known as lysozyme (the structure of which was ultimately determined by David Phillips and his colleagues).
- Later on, we will encounter enzymes that are similarly capable of breaking the covalent bonds in the backbones of DNA molecules.

# Thermodynamics

- Thermodynamics is the study of the effects of work, heat, and energy on a system.
- Thermodynamics is only concerned with macroscopic (large-scale) changes and observations

# Four Laws of Thermodynamics

## Zeroth law



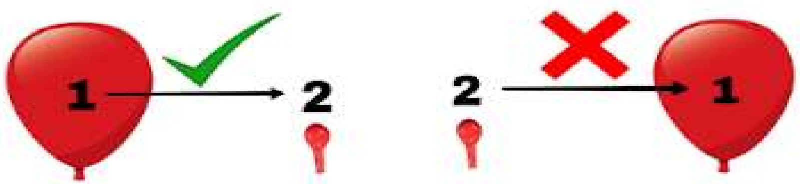

"If two bodies A and B are in thermal equilibrium with third body C, then body A and B are also in thermal equilibrium with each other"

## First law

$$\Delta E = Q - W$$

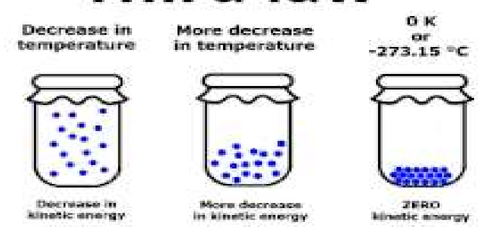
"The net change in total energy of a system ($\Delta E$) is equal to the heat added to the system ($Q$) minus work done by the system ($W$)"

## Second law




"In all the spontaneous processes, the entropy of the universe increases"

## Third law




"The value of entropy of a completely pure crystalline substance is zero at absolute zero temperature"